

**BODO, Norway**WMO: **011520**Lat: **67.267N**Lon: **14.359E**Elev: **42**StdP: **14.67**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 13.2       | 15.8 | 2.6                           | 6.3 | 15.7 | 5.2 | 7.2 | 17.6 | 43.1                  | 33.1 | 39.4 | 32.9 | 19.8                  | 90   | 0.823 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.6                    | 72.5            | 60.3 | 68.9 | 58.6 | 65.7 | 56.9 | 61.6                | 70.0 | 59.6 | 66.9 | 57.8 | 63.9 | 10.3                 | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 58.0                            | 71.9 | 64.1 | 56.4 | 67.8 | 62.0 | 54.9 | 64.2 | 60.3 | 27.5          | 69.9 | 26.2 | 66.9 | 25.0 | 63.9 | 71.1              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 36.7          | 32.1 | 28.7 | 10.4                       | 78.3 | 3.3                | 3.9 | 8.0                                                | 81.1 | 6.1        | 83.4 | 4.3        | 85.6 | 1.9        | 88.4 |     |     |
| (5)               |      |      | 8.9                        | 64.4 | 2.8                | 2.5 | 6.9                                                | 66.2 | 5.2        | 67.7 | 3.7        | 69.1 | 1.6        | 71.0 | (4) | (5) |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 41.9        | 30.6  | 29.9  | 32.1  | 38.2  | 45.2  | 51.1  | 56.4  | 55.8  | 50.5  | 42.4  | 36.2  | 33.1 | (6)  | (7)  |
|                                                                        | DBStd       | 11.23       | 7.13  | 7.32  | 6.19  | 5.51  | 5.86  | 5.64  | 5.55  | 4.94  | 5.16  | 5.90  | 6.40  | 7.11 | (7)  | (8)  |
|                                                                        | HDD50       | 3542        | 601   | 563   | 554   | 357   | 175   | 50    | 3     | 4     | 55    | 242   | 413   | 525  | (8)  | (9)  |
|                                                                        | HDD65       | 8455        | 1066  | 983   | 1019  | 805   | 614   | 419   | 274   | 288   | 434   | 700   | 863   | 990  | (9)  | (10) |
|                                                                        | CDD50       | 577         | 0     | 0     | 0     | 2     | 27    | 82    | 203   | 185   | 71    | 7     | 0     | 0    | (10) | (11) |
|                                                                        | CDD65       | 15          | 0     | 0     | 0     | 0     | 0     | 2     | 9     | 4     | 0     | 0     | 0     | 0    | (11) | (12) |
|                                                                        | CDH74       | 71          | 0     | 0     | 0     | 0     | 0     | 4     | 55    | 12    | 0     | 0     | 0     | 0    | (12) | (13) |
|                                                                        | CDH80       | 5           | 0     | 0     | 0     | 0     | 0     | 0     | 5     | 0     | 0     | 0     | 0     | 0    | (13) | (14) |
|                                                                        | WSAvg       | 14.2        | 18.7  | 18.4  | 16.4  | 13.3  | 12.0  | 10.5  | 9.2   | 10.0  | 12.7  | 15.2  | 16.9  | 17.7 | (14) | (15) |
|                                                                        | PrecAvg     | 41.7        | 3.5   | 2.8   | 2.9   | 2.4   | 2.1   | 2.4   | 3.2   | 3.4   | 4.8   | 5.6   | 4.2   | 4.3  | (15) | (16) |
| (16) Precipitation                                                     | PrecMax     | 60.2        | 9.7   | 6.6   | 7.6   | 6.1   | 5.0   | 5.4   | 7.5   | 9.4   | 10.8  | 11.1  | 9.5   | 10.2 | (16) | (17) |
|                                                                        | PrecMin     | 26.4        | 0.0   | 0.1   | 0.4   | 0.6   | 0.4   | 0.4   | 0.4   | 0.6   | 0.7   | 1.2   | 0.0   | 0.2  | (17) | (18) |
|                                                                        | PrecStd     | 8.4         | 2.1   | 1.3   | 1.5   | 1.3   | 1.2   | 1.1   | 1.7   | 2.2   | 2.6   | 2.4   | 2.1   | 2.2  | (18) | (19) |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |      | (19) | (20) |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 45.7  | 44.7  | 46.5  | 57.3  | 68.6  | 73.4  | 79.6  | 75.4  | 66.4  | 58.3  | 50.9  | 47.5 | (20) | (21) |
|                                                                        |             | MCWB        | 42.6  | 41.8  | 40.5  | 46.4  | 54.8  | 59.5  | 64.4  | 62.0  | 57.0  | 50.7  | 46.9  | 44.4 | (21) | (22) |
|                                                                        | 2%          | DB          | 43.8  | 42.8  | 43.7  | 52.1  | 62.7  | 68.4  | 74.4  | 70.9  | 62.7  | 55.0  | 48.4  | 45.7 | (22) | (23) |
|                                                                        |             | MCWB        | 40.6  | 40.3  | 40.1  | 43.6  | 51.5  | 56.4  | 61.3  | 60.4  | 54.6  | 48.6  | 45.1  | 42.8 | (23) | (24) |
|                                                                        | 5%          | DB          | 41.8  | 41.3  | 42.2  | 48.6  | 58.4  | 64.4  | 69.8  | 67.3  | 60.2  | 52.2  | 46.7  | 44.1 | (24) | (25) |
|                                                                        |             | MCWB        | 39.1  | 38.8  | 38.9  | 41.8  | 49.4  | 54.4  | 59.5  | 58.7  | 53.5  | 47.7  | 43.0  | 41.3 | (25) | (26) |
|                                                                        | 10%         | DB          | 39.8  | 39.5  | 40.6  | 45.5  | 54.1  | 60.7  | 66.2  | 64.0  | 57.6  | 50.2  | 44.9  | 42.4 | (26) | (27) |
|                                                                        |             | MCWB        | 37.0  | 36.8  | 37.6  | 40.5  | 46.6  | 52.9  | 58.0  | 56.7  | 52.3  | 46.5  | 41.2  | 39.4 | (27) | (28) |
|                                                                        | 0.4%        | WB          | 43.9  | 43.0  | 43.0  | 47.2  | 55.5  | 60.3  | 66.2  | 63.7  | 58.2  | 51.9  | 48.1  | 46.1 | (28) | (29) |
|                                                                        |             | MCDB        | 44.6  | 43.8  | 44.3  | 56.5  | 65.9  | 70.9  | 76.7  | 72.9  | 64.5  | 55.7  | 49.6  | 46.7 | (29) | (30) |
| (30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB          | 41.3  | 41.0  | 41.1  | 44.5  | 52.4  | 57.3  | 62.4  | 61.2  | 56.0  | 50.1  | 45.9  | 43.7 | (30) | (31) |
|                                                                        |             | MCDB        | 42.7  | 42.2  | 42.9  | 49.5  | 61.3  | 66.5  | 71.2  | 69.0  | 60.3  | 53.4  | 47.9  | 44.8 | (31) | (32) |
|                                                                        | 5%          | WB          | 39.3  | 39.0  | 39.4  | 43.0  | 49.8  | 55.3  | 60.4  | 59.3  | 54.7  | 48.6  | 43.7  | 41.6 | (32) | (33) |
|                                                                        |             | MCDB        | 41.1  | 40.7  | 41.5  | 46.6  | 57.1  | 62.7  | 68.3  | 65.6  | 58.5  | 51.5  | 46.1  | 43.5 | (33) | (34) |
|                                                                        | 10%         | WB          | 37.2  | 37.1  | 37.7  | 41.4  | 47.5  | 53.4  | 58.5  | 57.6  | 53.1  | 46.9  | 41.5  | 39.5 | (34) | (35) |
|                                                                        |             | MCDB        | 39.5  | 39.2  | 40.1  | 44.8  | 53.1  | 59.2  | 64.9  | 62.8  | 56.9  | 49.9  | 44.3  | 41.7 | (35) | (36) |
|                                                                        | 5% DB       | MDBR        | 7.0   | 7.1   | 7.3   | 8.1   | 9.0   | 8.9   | 9.6   | 9.3   | 8.2   | 6.5   | 6.3   | 6.8  | (36) | (37) |
|                                                                        |             | MCDBR       | 8.1   | 7.8   | 8.0   | 11.3  | 15.0  | 15.9  | 17.1  | 15.6  | 11.4  | 7.6   | 7.3   | 7.7  | (37) | (38) |
|                                                                        |             | MCWBR       | 7.5   | 7.5   | 6.5   | 6.8   | 7.5   | 7.9   | 7.9   | 7.6   | 6.1   | 5.5   | 6.3   | 7.5  | (38) | (39) |
|                                                                        |             | MCWBR       | 8.1   | 7.8   | 7.7   | 9.2   | 13.9  | 14.1  | 15.7  | 14.0  | 9.8   | 7.0   | 7.5   | 7.9  | (39) | (40) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.365 | 0.280 | 0.313 | 0.334 | 0.340 | 0.339 | 0.347 | 0.336 | 0.320 | 0.289 | 0.349 | N/A  | (40) | (41) |
|                                                                        |             | taud        | 2.639 | 2.176 | 2.272 | 2.352 | 2.427 | 2.486 | 2.505 | 2.526 | 2.539 | 2.312 | 2.596 | N/A  | (41) | (42) |
|                                                                        | Ebn at Noon |             | 56    | 171   | 232   | 261   | 272   | 275   | 269   | 260   | 236   | 169   | 50    | 0    | (42) | (43) |
|                                                                        |             | Edh at Noon | 5     | 15    | 24    | 29    | 30    | 29    | 28    | 24    | 19    | 13    | 4     | 0    | (43) | (44) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 14    | 118   | 477   | 1012  | 1425  | 1562  | 1426  | 1065  | 565   | 199   | 33    | 0    | (44) | (45) |
|                                                                        | RadStd      |             | 2     | 22    | 71    | 99    | 101   | 189   | 165   | 117   | 63    | 40    | 3     | 0    | (45) | (46) |

**Historical Trends**

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | +0.61 | N/A     | N/A    | N/A     | N/A   | N/A   | -167        | -239  | N/A   | N/A   |
| (47) | Regional (24 neighbors) | +0.74 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page